

Vision-Language Models with In-Context Learning for In-Camera Portrait Composition Guidance

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Abstract

Capturing aesthetically pleasing portraits requires specialized compositional knowledge that ordinary users often lack, making it difficult for them to achieve optimal camera framing. While vision-language models (VLMs) excel at general image understanding, they frequently produce overly positive feedback and lack the granular structural awareness needed to generate actionable compositional guidance. To address this, we present a visual in-context learning (ICL) framework that equips VLMs with augmented aesthetic evaluation capabilities. By retrieving structurally similar, professionally reframed image-composition analyses as visual prompts,

our framework grounds VLM's reasoning in established aesthetic standards, e.g., the rule of thirds and leading lines, and enables mapping compositional flaws to specific camera instructions.

CCS Concepts

• **Human-centered computing**; • **Computing methodologies**
→ **Visual content-based indexing and retrieval**;

Keywords

Image composition, photography guidance, language models

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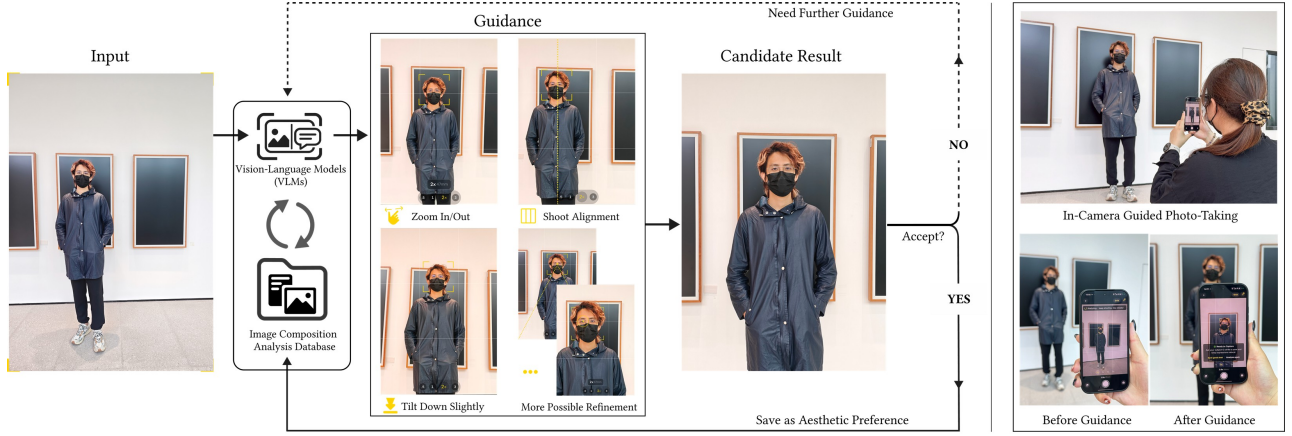


Figure 1: Our framework facilitates portrait composition refinement by translating composition principles (e.g., symmetry and negative space) into actionable camera guidance. Leveraging composition analysis examples, VLMs generate camera instructions and iteratively update them when further refinement is needed.

1 Introduction

Smartphone photography is a routine medium for documenting daily life, yet portrait composition remains difficult for non-experts: small changes in subject position, viewpoint, and background can alter visual balance [5, 12, 16]. Prior capture-time tools explore adaptive overlays [3, 7, 9], while cropping improves images after the shutter is pressed [6, 8]. Users still need situated guidance that converts composition principles into executable camera actions.

Recent developments enable large models to reason over images and videos [4, 10, 11], but visual ICL depends strongly on the supplied domain knowledge [1, 19]. For portrait composition, effective demonstrations must encode subject geometry, aesthetic principles, and camera operations. As Figure 1 shows, we explore this enhancement of VLMs in aesthetics and spatial understanding via: (1) a retrieval-augmented visual ICL framework; (2) 5,068 expert-rectified composition analyses from representative open-source datasetss [2, 13, 17, 18]; and (3) a mobile camera prototype that presents inspiring in-situ guidance. These components probe agentic capture workflows that support users’ creative confidence.

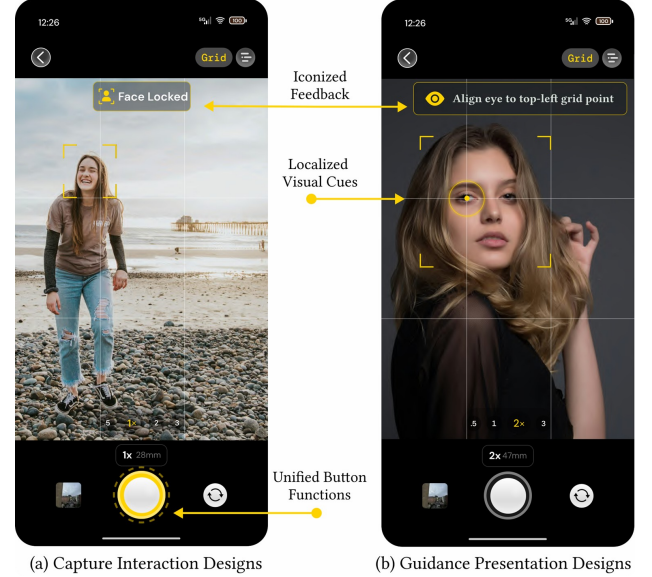


Figure 3: UI designs for composition. Photos on Unsplash [14]

Examples of the Structured Image Analysis

A photograph of a woman in a green dress standing on a staircase. The image is overlaid with a yellow grid and various geometric shapes (circles, rectangles) indicating analysis points. A red dot marks the centroid, and a yellow rectangle marks the bounding box.

Geometric Understanding

- Centroid: (0.3, 0.15)
- Bounding Box: [0.17, 0.03, 0.79, 1.0], [0.24, 0.03, 0.38, 0.24]
- Zoom: 1.5x

Composition Principles

- rule_of_thirds: 0.90 (*Dominant*); leading_lines: 0.70; symmetric: 0.10
- Framing: near waist_up

Image Evaluation

- Aesthetic Scores for Composition: 0.88
- Composition Binary Classification: Good

Camera Instructions

- Use railings as **leading lines** toward subject
- Position the subject's face on the **upper-left intersection of the thirds grid**

A photograph of a woman in a red dress standing in a doorway. The image is overlaid with a yellow grid and various geometric shapes (circles, rectangles) indicating analysis points. A red dot marks the centroid, and a yellow rectangle marks the bounding box.

Geometric Understanding

- Centroid: (0.63, 0.23)
- Bounding Box: [0.54, 0.15, 0.73, 0.99], [0.59, 0.16, 0.74, 0.31]
- Zoom: 1x

Composition Principles

- rule_of_thirds: 0.95 (*Dominant*); symmetric: 0.65; centering: 0.10
- Framing: near half_body

Image Evaluation

- Aesthetic Scores for Composition: 0.92
- Composition Binary Classification: Good

Camera Instructions

- Use arch to create a **frame within a frame**
- Leave negative space to include more background elements
- Use bokeh to elevate the portrait

Figure 2: ICL Examples in composition analysis database.

2 ICL for Composition Guidance

We started by interviewing eight expert photographers and distilled four design goals: guidance should be situated, glanceable, inspirational, and execution-centric. Our generated guidance uses brief cues such as “center subject,” “zoom in,” or “tilt camera up” instead of dense explanations or ghost overlays. Our framework achieves these via retrieval-augmented visual ICL. As shown in Figure 2, each curated composition analysis pairs an image with structured descriptions covering subject geometry, composition principles, aesthetic assessment, and executable guidance. At the inference stage, our framework extracts geometric and semantic cues from the live frame and retrieves structurally similar examples as professional contexts, enabling the VLM to generate a small set of

actionable camera adjustments. We implemented this ICL pipeline with Qwen3.6-flash on a Samsung Galaxy S25: a tap captures a photo, while a long press requests analysis and returns brief text, icons, and localized visual emphasis (Figure 3).

3 Preliminary Evaluation and Discussion

We conducted an in-situ study with 8 participants to examine how retrieval-grounded guidance affects capture-time composition. Participants completed five portrait tasks under three conditions: Static Grid, Zero-shot VLM Guidance, and our ICL-enhanced guidance (Figure 4). Post-interviews suggest that the system inspired users' composition process: 6/8 participants reported considering alternative framings, while expert users (2/8) used the guidance as a secondary check rather than a directive. Figure 5 indicates that our guidance formulation can extend to multi-person, low-light, non-human, and varied-aspect-ratio scenes to benefit photo taking.

In conclusion, our work highlights the potential of visual ICL to support aesthetic decision-making and agentic systems that provide context-aware and inspiring support for everyday visual creation.



Figure 4: Participant-produced images from the user study. Aesthetic ratings are computed with Q-Align [15].

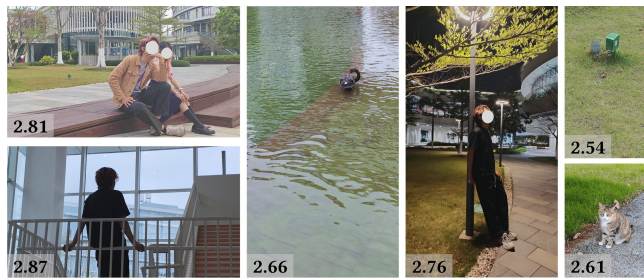


Figure 5: Composition inspiration across diverse scenarios. Overlaid values indicate Clip-A [13] aesthetic scores.

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